Rules and Numbers

Cut along the dashed lines, then fold each strip down the middle so the prompt and the answer are back to back.

Small UA weight limit	Less than 55 pounds at takeoff, counting everything on board or attached.
Maximum groundspeed	87 knots (100 miles per hour).
Maximum altitude	400 feet AGL. Within a 400-foot radius of a structure, up to 400 feet above its top.
Minimum flight visibility	3 statute miles, measured from the control station.
Cloud clearance	At least 500 feet below clouds and 2,000 feet horizontally from them.
Accident report deadline	Within 10 calendar days, through FAADroneZone.
What makes an accident reportable?	Serious injury, any loss of consciousness, or damage to other property unless repair is $500 or less, or a total loss is worth $500 or less.
Does damage to the drone itself count?	No. The property trigger covers property other than the small UA.

Alcohol: hours from drinking to flying	8 hours.
Alcohol: blood alcohol limit	Below 0.04, and you may not fly while under the influence at all.
Minimum age to hold the certificate	16.
Minimum age to take the knowledge test	14.
Knowledge test format	60 scored questions, 3 options each, 120 minutes, 70 percent (42 correct) to pass.
Wait after failing the test	14 days.
How long a passing test result stays valid	24 calendar months for applying.
Temporary certificate	Valid for up to 120 calendar days.
Recurrent training interval	Every 24 calendar months, ending on the last day of the month.

ALC-677	The free online recurrent course for remote pilots.
ALC-515	The recurrent course for Part 61 pilots with a current flight review.
Does the certificate expire?	No. Recency lapses, and recurrent training restores it.
Address change deadline	30 days.
Application form	FAA Form 8710-13, completed in IACRA.
Part 107 registration	Each drone separately, $5, valid for 3 years, through FAADroneZone.
Recreational registration	One number for all drones you own. Drones of 0.55 pounds or less are exempt.
Who may ask for your certificate?	The FAA, NTSB, TSA, and any federal, state, or local law enforcement officer.
Who may fly without a certificate?	A person under the direct supervision of a remote PIC who can take over immediately.

Who has final authority over a flight?	The remote PIC.
Emergency deviation	You may deviate only as far as the emergency requires, and send a written report if the FAA asks.
Night flight needs	Current training or test after April 6, 2021, and anti-collision lighting visible for 3 statute miles.
Civil twilight (outside Alaska)	The 30 minutes before sunrise and the 30 minutes after sunset. Lighting is required.
Visual observer	Optional. Needs no certificate. Must keep effective communication with the crew.
How many aircraft at once?	One, unless you have a waiver.
Hazardous material	Prohibited, and the rule cannot be waived.
Dropping objects	Allowed if it creates no undue hazard to people or property.
Flying from a moving vehicle	Never from an aircraft. From land or water vehicles only over a sparsely populated area, and not carrying others' property for hire.

Right of way	The small UA yields to all aircraft and may pass only when well clear.
Paid cargo limits	Under 55 pounds in total, within visual line of sight, within one state, no hazardous material, not from a moving vehicle.
Category 1	0.55 pounds or less, with no exposed rotating parts that could lacerate skin.
Category 2	Declaration of Compliance and a label. Injury limit of 11 foot-pounds.
Category 3	Declaration of Compliance and a label. Injury limit of 25 foot-pounds. No open-air assemblies.
Category 4	Airworthiness certificate under Part 21, and flight manual limits.
Remote ID paths	Standard Remote ID, a broadcast module, or flying inside a FRIA.
Broadcast stops in flight	Land as soon as practicable.
Who can apply for a FRIA?	FAA-recognized community-based organizations and educational institutions.

Drug or alcohol conviction	Denial of an application for up to 1 year, or suspension or revocation of the certificate.
Waivable rule examples	Night, visual line of sight, visual observer, one aircraft, right of way (107.37(a)), over people, airspace, operating limits.
Rules that cannot be waived	Examples are hazardous material (107.36), alcohol and drugs (107.27), and accident reporting (107.9).

Airspace and Charts

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Authorization is needed in	Class B, Class C, Class D, and the surface area of Class E designated for an airport.
Airspace with no authorization needed	Class G and other Class E.
Class A begins at	18,000 feet MSL.
Typical Class B ceiling	About 10,000 feet MSL.
Class B on a chart	Solid blue rings, like an upside-down wedding cake.
Class C on a chart	Solid magenta rings.
Class D on a chart	A dashed blue circle, with the ceiling in a dashed box.
Class E to the surface on a chart	A dashed magenta line.

Faded magenta band	Class E begins at 700 feet AGL.
Faded blue band	Class E begins at 1,200 feet AGL.
Class E with no other floor shown	Begins at 14,500 feet MSL.
Class D typical ceiling	About 2,500 feet above the airport.
Class C typical shape	Inner core to about 5 nautical miles from the surface. Outer shelf to about 10 nautical miles from about 1,200 feet above the airport.
Fraction 100 over 25	Ceiling 10,000 feet MSL, floor 2,500 feet MSL.
SFC in a fraction	The floor is the surface.
Dashed box with 33	Class D ceiling of 3,300 feet MSL.
Obstacle label 1865 (998)	The top is 1,865 feet MSL. The height is 998 feet AGL.

Ground elevation from an obstacle label	Subtract the parentheses from the bold number.
Airport data block: CT	Control tower, followed by its frequency.
Star or asterisk after a tower frequency	A part-time tower.
Circled C (plain C in the book)	CTAF, the Common Traffic Advisory Frequency.
First number on the airport data line	Field elevation in feet MSL.
L in an airport data line	Runway lighting. An asterisk after the L means pilot-controlled or limited.
72 in an airport data line	Longest runway of 7,200 feet.
Chart scale and cycle	Sectional scale is 1:500,000. Revised on a 56-day cycle.
Tower closed at Class D	The airspace may become Class E or Class G. Check the Chart Supplement.

Restricted area	Get permission from the using or controlling agency.
Prohibited area	Closed for security or national welfare reasons. No flight without permission from the using or controlling agency, and do not expect to get it.
Military Operations Area	Military training. No permission needed, but use caution.
Alert area	High volume of pilot training or unusual activity. Be alert.
Warning area	Beyond 3 nautical miles from the coast, with possibly hazardous activity.
National Security Area	Voluntary avoidance, which a NOTAM can make temporarily prohibited.
Military training route numbers	Four digits: at or below 1,500 feet AGL. Three digits: goes above 1,500 feet AGL.
Stadium restriction	Within 3 nautical miles and up to 3,000 feet AGL, from 1 hour before to 1 hour after the event.
LAANC	Near-real-time authorization up to the grid altitude, requested up to 90 days ahead.

LAANC grid shows 0	No automatic approval. Request through FAADroneZone.
B4UFLY	Information only. It does not issue authorizations.

METAR and TAF Codes

Cut along the dashed lines, then fold each strip down the middle so the prompt and the answer are back to back.

METAR	A routine weather report, normally once an hour.
SPECI	A special weather report issued between routine reports.
TAF	A terminal aerodrome forecast.
Z in a time	Zulu, or UTC.
18012G22KT	Wind from 180 degrees at 12 knots, gusting to 22 knots.
VRB	Variable wind direction.
10SM	Visibility of 10 statute miles.
P6SM	Visibility of more than 6 statute miles, in a TAF.

FEW	1 to 2 eighths of the sky covered.
SCT	3 to 4 eighths of the sky covered.
BKN	5 to 7 eighths of the sky covered. Counts as a ceiling.
OVC	8 eighths of the sky covered. Counts as a ceiling.
BKN012	A broken layer at 1,200 feet AGL.
VV004	Sky obscured, with a vertical visibility of 400 feet. Counts as a ceiling.
CB after a cloud layer	Cumulonimbus, a thunderstorm cloud.
RA	Rain.
SN	Snow.

BR	Mist.
FG	Fog.
TS	Thunderstorm.
- before a weather code	Light.
+ before a weather code	Heavy.
VC	In the vicinity.
SH	Showers.
13/12	Temperature of 13°C and dew point of 12°C.
A3002	Altimeter setting of 30.02 inches of mercury.

RMK	The remarks section begins.
FM061800	From the 6th at 1800Z, with the forecast changing from then on.
TEMPO	A temporary change within the stated period.
BECMG	A gradual change over a stated period.
PROB30	A 30 percent probability in the stated period.
AIRMET Sierra	IFR conditions and mountain obscuration.
AIRMET Tango	Turbulence, strong surface wind, and low-level wind shear.
AIRMET Zulu	Icing and freezing level.
Convective SIGMET	Thunderstorm hazards, valid for up to 2 hours.

Airport and Radio

Cut along the dashed lines, then fold each strip down the middle so the prompt and the answer are back to back.

CTAF	The shared frequency at a non-towered airport, or at a towered one when the tower is closed.
UNICOM	The airport's non-government advisory radio.
ATIS	A continuous recording of airport weather and information.
Standard traffic pattern turns	Left, unless shown otherwise.
Downwind leg	Parallel to the runway, opposite to the landing direction.
Base leg	At a right angle to the runway.
Typical pattern altitude	About 1,000 feet above the ground.
Danger zones near a runway	Off both ends, where arriving and departing aircraft are lowest.

Runway number 18	About 180 degrees magnetic.
Runway marking color	White.
X on a runway or taxiway	Closed.
Displaced threshold	Usable for takeoff and rollout, but not for touchdown.
Wind cone	Shows wind direction.
Traffic pattern indicators	Show which way aircraft turn in the pattern.
White and green beacon	A civilian land airport.
White and yellow beacon	A water airport.
Two quick white flashes between green	A military airport.

Roger	"I received your message."
Wilco	"I will comply."
Traffic two o'clock	Ahead and to the right.
The number 9 on the radio	Niner.
Phonetic alphabet: X	X-ray.

Mnemonics and Human Factors

Cut along the dashed lines, then fold each strip down the middle so the prompt and the answer are back to back.

IMSAFE	Illness, Medication, Stress, Alcohol, Fatigue, Emotion.
PAVE	Pilot, Aircraft, enVironment, External pressures.
DECIDE	Detect, Estimate, Choose, Identify, Do, Evaluate.
5P check	Plan, Plane, Pilot, Passengers, Programming.
3P model	Perceive, Process, Perform.
Anti-authority antidote	"Follow the rules. They are usually right."
Impulsivity antidote	"Not so fast. Think first."
Invulnerability antidote	"It could happen to me."

Macho antidote	"Taking chances is foolish."
Resignation antidote	"I'm not helpless. I can make a difference."
Dark adaptation	About 30 minutes for the eyes to adjust fully.
Best way to see at night	Look a little to one side of the object.
Autokinesis	A stationary light seems to move when you stare at it in the dark.
Empty-field myopia	In a featureless sky, your eyes relax to a short focus.
Tingling and dizziness under stress	Hyperventilation.
Acute and chronic fatigue	Acute is short-term and fixed by rest. Chronic builds up and takes longer to recover from.
Headache and tiredness in the heat	Dehydration. Stop, drink water, and find shade.

CRM closed-loop communication	The message is repeated back and confirmed.

Loading and Performance

Cut along the dashed lines, then fold each strip down the middle so the prompt and the answer are back to back.

Load factor at 30 degrees of bank	1.15.
Load factor at 45 degrees of bank	1.41.
Load factor at 60 degrees of bank	2.0.
Standard atmosphere at sea level	29.92 inHg and 15°C (59°F).
Standard lapse rate	About 2°C per 1,000 feet.
Three H's of high density altitude	High, hot, and humid.
Effect of high density altitude	Thinner air and reduced performance.
Is humid air heavier?	No. Water vapor is lighter than dry air, so humid air is less dense.

Cloud base estimate	Temperature and dew point spread in °F, divided by 4.4, times 1,000 feet.
Stall speed in a turn	Rises with the square root of the load factor.
Forward center of gravity on a multirotor	The heavy-side motors work harder, which costs endurance and margin.
Lithium battery storage	About half charge, cool and dry, or as the manufacturer directs.